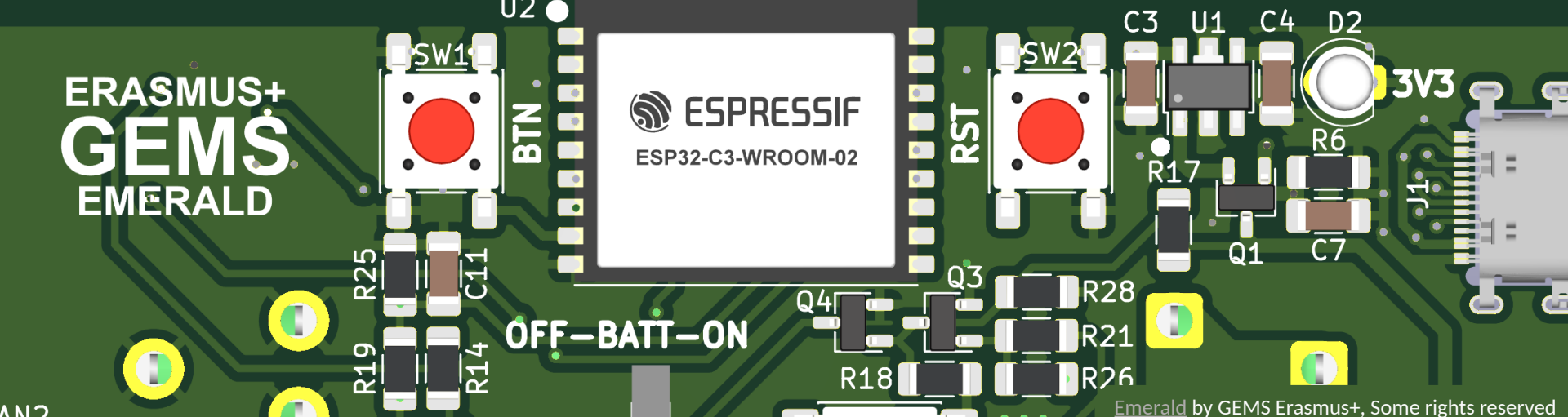


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Wirelessly controlled robot – User interface

Building Robots: System Integration

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About

- 1 User interface
- 2 Implementation
- 3 Web and native application
- 4 Cloud and edge application

User interface





Level of complexity

Graphical user interface (GUI)

1

- complex software solution
- flexible, powerful

2

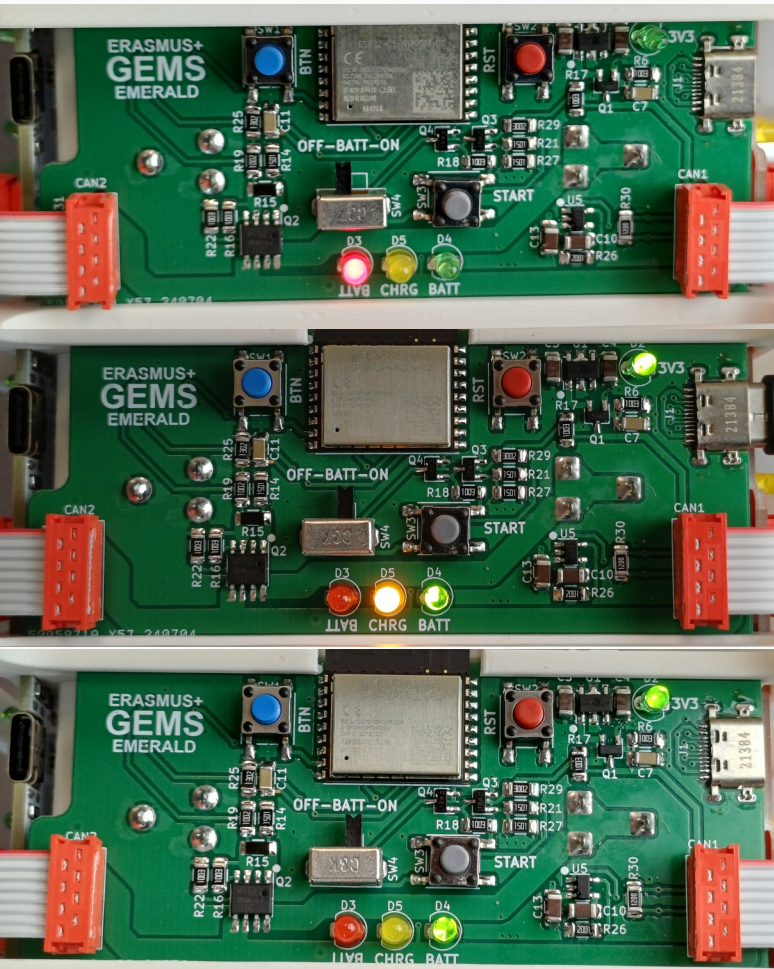
- LED, Button, switch → state
- simple hardware solution

3

- Mix use based on information level
- low level → hardware interface

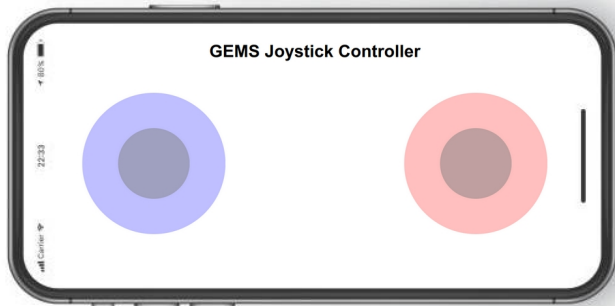
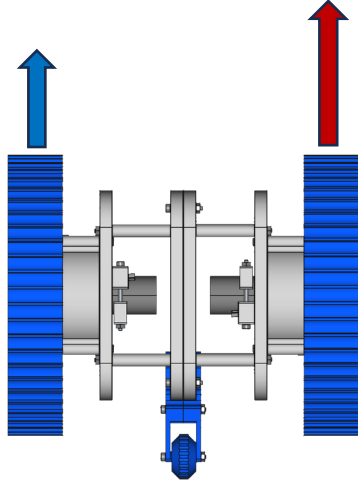
Implementation





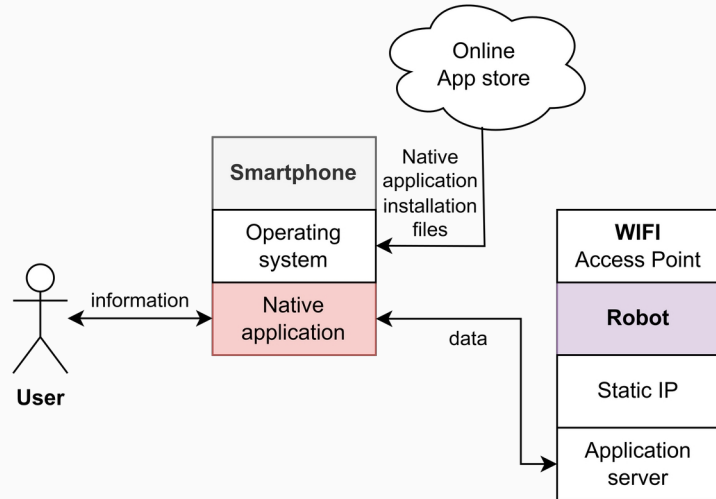
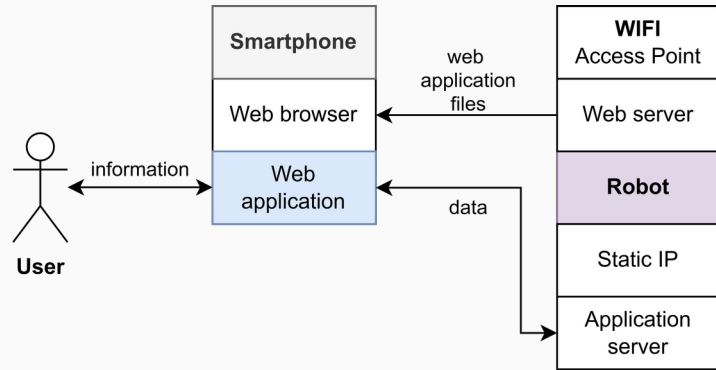
Use of indicators

- 1 Information about system state
 - Error (battery polarity)
 - Warning (battery charging)
 - OK (power supply status)
- 2
- 3 Important for system diagnosis
 - when GUI doesn't work



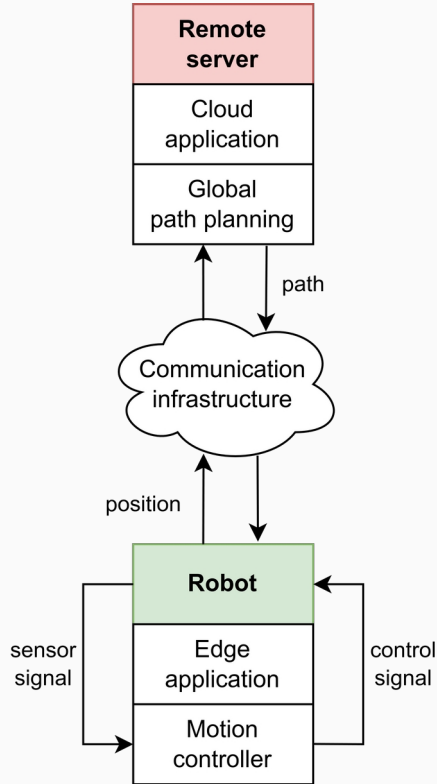
Use of a smartphone

- 1 Ubiquitous
→ no need for special hardware
- 2 Touchscreen
→ intuitive interaction
- 3 Flexible
→ design can be easily changed
→ use of existing technology



Web vs. native application

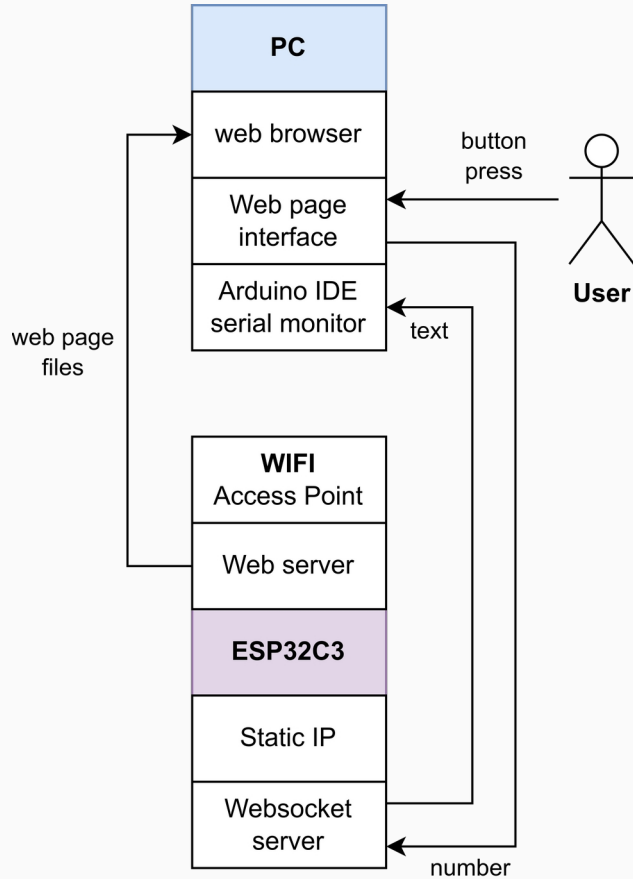
- 1 Native application
→ designed for specific OS
- 2 Web app/page
→ designed for a browser
- 3 Selection depends on a use case
 - education use → web application
 - doesn't require installation



Cloud vs. edge application

- 1 Cloud application remote location → communication
- 2 Edge application responsive → less resources
- 3 Edge for immediate control Cloud for intensive computation

Task



Web user interface

Web page and Arduino program

- 1 ESP32C3/S3, browser
Web server on microcontroller
- 2 Web page with 3 control buttons
→ send a number
- 3 Websocket server
→ act based on the message

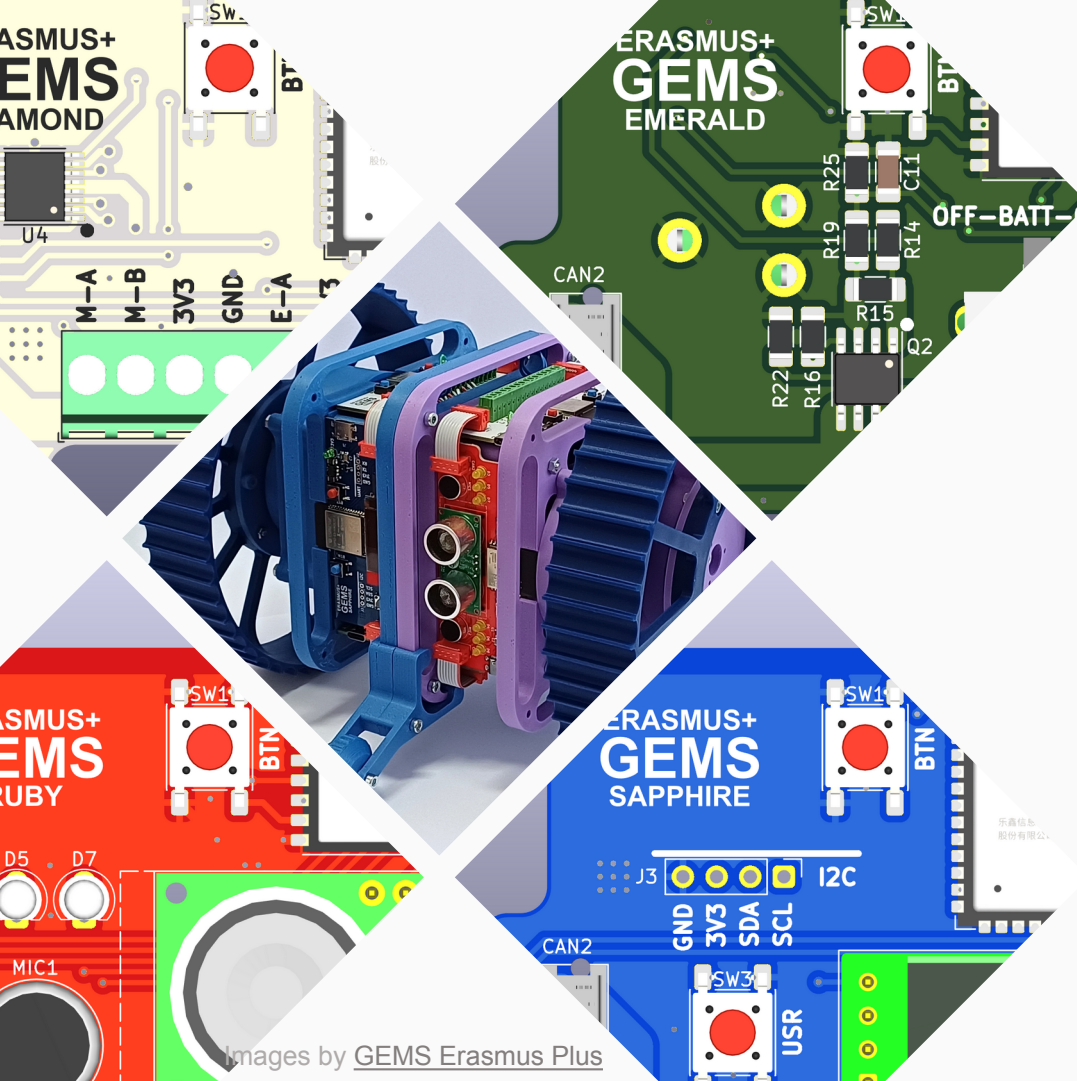
Conclusion





Human machine interface

- 1 Different level of complexity indicator vs. touchscreen app
- 2 Many different possibilities
- 3 Simplicity and functionality



Thank you for watching!

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